Chapter 3: Formalism

- In quantum mechanics we study:

I) States: which lives in a Hilbert space

II) obserables: linear operator on Hilbert space

- Hilbert Space: is an inner product vector space.

in QM, we will have two Hilbert Spaces:

1) $\mathbb{C}^n = \left\{ \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} : v_i \in \mathbb{C} \right\}$

2) Square-integrable functions $L^2[a,b]$

- the set of all functions such that

$$\int_a^b \underbrace{|f|^2}\, dx < \infty \;. \Rightarrow \; |f|^2 \to 0 \text{ as } \to \pm\infty$$

Observables:

let's say we a Measurable quantity call in q.

with the corresponding operator $\hat{Q}$.

The expection value of q,

$3+5i \neq 3+5i^*$

$$\langle \hat{Q} \rangle \equiv \langle \psi | \hat{Q} \psi \rangle$$ where $\psi \in \mathcal{H}$.

$i \neq i^*$

$x = x^*$
$5 = 5^*$

Exp 1: Start from the fact $\langle \hat{Q} \rangle = \langle \hat{Q} \rangle^*$

Show that $\hat{Q}$ must be Hermitian.

$$\langle \hat{Q} \rangle = \langle \psi | \hat{Q} \psi \rangle$$

$$\langle \hat{Q} \rangle^* = \langle \psi | \hat{Q} \psi \rangle^* = \langle \hat{Q} \psi | \psi \rangle$$

$\Rightarrow \langle \psi | \hat{Q} \psi \rangle = \langle \hat{Q} \psi | \psi \rangle$, for all $\psi \in \mathcal{H}$.

$\Rightarrow \boxed{\langle f | \hat{Q} g \rangle = \langle \hat{Q} f | g \rangle}$ for any $f, g \in \mathcal{H}$

$$\Rightarrow \boxed{\hat{Q} = \hat{Q}^\dagger}$$

• Special States

Say that we a state ϕ such that if we prepare our experiment at ϕ, and measure q we get q_0.

Means $\langle \hat{Q} \rangle = q_0$ and $\sigma^2 = 0$.

but $\sigma^2 = \langle \hat{Q}^2 \rangle - \langle \hat{Q} \rangle^2$

$\langle \psi | \hat{T} \hat{T} f \rangle$
$\langle \hat{T} \psi | \hat{T} f \rangle$

notice

$$\langle \hat{Q}^2 \rangle - \langle \hat{Q} \rangle^2 = \langle \hat{Q}^2 - 2\hat{Q}\langle \hat{Q} \rangle + \langle \hat{Q} \rangle^2 \rangle$$

$$= \langle (\hat{Q} - \langle \hat{Q} \rangle)^2 \rangle = \langle \psi | (\hat{Q} - q_0)^2 \psi \rangle$$

• using the fact $(\hat{Q} - q_0)$ is Hermitian.

$\langle (\hat{Q} - q_0) f | g \rangle = \langle \hat{Q} f - q_0 f | g \rangle = \langle \hat{Q} f | g \rangle - \langle q_0 f | g \rangle$
$= \langle f | \hat{Q} g \rangle - \langle f | q_0 g \rangle = \langle f | (\hat{Q} - q_0) g \rangle$

$$\sigma^2 = \langle (\hat{Q} - q_0)\phi | (\hat{Q} - q_0)\phi \rangle = 0$$

• use $\langle v | v \rangle = 0 \Longleftrightarrow v = 0$

$\Rightarrow \quad (\hat{Q} - q_0)\,\phi = 0$

$\Rightarrow \quad \hat{Q}\,\phi = q_0\,\phi$: q_0 eigenvalue of $\hat{Q}$

So ϕ is an eigenfunction of $\hat{Q}$!

$\boxed{\text{• The States with deterministic measurement of } q \text{ are the eigenfunction of } \hat{Q}.}$

• Fact: these States (the eigenfunction of $\hat{Q}$) are complete. (Span $\mathcal{H}$).

So If the eigenfunctions are $\{\phi_1, \phi_2 \dots \phi_n\}$, then any $f \in \mathcal{H}$,

$$f = \sum \alpha_i \phi_i \quad \text{or} \quad f = \int \alpha\,\phi$$

For any observable q with the Corresponding Herimitian operator $\hat{Q}$ such that:

1) $\hat{Q}$ has real eigenvalues

2) the eigenfunctions are complete. of $\hat{Q}$

3) $\phi_1, \phi_2, \alpha, \beta$

$\langle \phi_1 | \phi_2 \rangle = 0$

• The Spectrum of a Hermitian Operator

the spectrum of an operator is the set of all the eigenvalue of it.
all possible values that we can measure.

• Discrete Spectra:

* the eigenfunctions of such operator have a physical meaning.

• Continuous Spectra

* the eigenfunctions are non-normalizable

$\Rightarrow$ has no physical meaning.

• orthogonality for f_λ and $f_{\lambda'}$

$$\langle f_\lambda | f_{\lambda'} \rangle = \delta(\lambda - \lambda') = \begin{cases} \infty ; & \lambda = \lambda' \\ 0 ; & \lambda \neq \lambda' \end{cases}$$

Exp : Show that $\hat{x}$, $\hat{p}$ are: Hint $\int_{-\infty}^{\infty} e^{-iq\frac{x}{\hbar}}\,dx = 2\pi\hbar\,\delta(q)$.

a) Hermitian

b) have continuous spectra

let take $\hat{x}$: $\underline{T(f)} = hf$

a) $\langle f | \hat{x} g\rangle = \int_{-\infty}^{\infty} f^*(x)\, x\, g(x)\,dx$

$= \int_{-\infty}^{\infty} x^* f^*(x)\, g(x)\,dx = \int_{-\infty}^{\infty} (x f(x))^*\, g(x)\,dx$

$= \langle xf|g\rangle = \langle \hat{x} f|g\rangle$.

let λ be an eigenvalue of $\hat{x}$

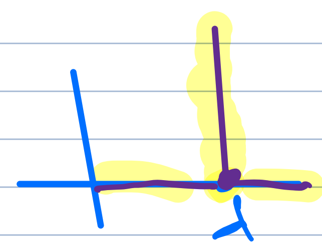
$\hat{x}\,\psi(x) = \lambda\,\psi(x)$ $(\psi \neq 0)$

$x\psi(x) = \lambda\,\psi(x) \Rightarrow \underbrace{(x-\lambda)}\,\psi(x) = 0$

if $x \neq \lambda \Rightarrow \boxed{\psi(x) = 0}$ $\lambda = 0$: $0 \cdot \psi(x) = 0$

at $x=\lambda$, $\psi(x) \neq 0$.

$\psi(x) = \delta(x-\lambda) = \begin{cases} \infty; & \lambda = x \\ 0; & x \neq \lambda \end{cases}$



$\lambda \in \mathbb{R}$.

any $f(x) \in \mathcal{H}$.

$$\boxed{f(x) = \int_{-\infty}^{\infty} c(\lambda)\, \delta(x-\lambda)\, d\lambda}$$

For $\hat{p}$:

Part a)

$\langle \hat{p} f | g \rangle = \langle f | \hat{p} g \rangle$.

$$\langle \hat{p} f | g \rangle = \int_{-\infty}^{\infty} \left(-i\hbar \frac{d}{dx} f\right)^* g \; dx$$

$$= i\hbar \int_{-\infty}^{\infty} \frac{df^*}{dx} g \, dx = i\hbar \left[\cancel{f^* g}\Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} f^* \frac{dg}{dx} dx \right]$$

$du = \frac{df^*}{dx}$ $\quad v = g$

$u = f^*$ $\quad dv = \frac{dg}{dx}$

$$= -i\hbar \int_{-\infty}^{\infty} f^* \frac{dg}{dx} dx$$

$$= \int_{-\infty}^{\infty} f^* \left(-i\hbar \frac{d}{dx}\right) f$$

$= \langle f | \hat{p} g \rangle$.

$\hat{p}\, \psi(x) = \lambda\, \psi(x)$

$$-i\hbar \frac{d\psi}{dx} = \lambda \psi$$

$$\frac{d\psi}{\psi} = \frac{i}{\hbar} \lambda \, dx \Rightarrow \ln \psi = \frac{i}{\hbar} \lambda x + C_1$$

$$\Rightarrow \quad \psi = A \, e^{\frac{i}{\hbar} \lambda x}$$

$$\psi_\lambda(x) = A e^{\frac{i}{\hbar} \lambda x} \quad . \quad \psi_{\lambda'} = A e^{\frac{i}{\hbar} \lambda' x}$$

$$\langle \psi_\lambda | \psi_{\lambda'} \rangle = \int_{-\infty}^{\infty} |A|^2 \, e^{-\frac{i}{\hbar} \lambda x} \cdot e^{\frac{i}{\hbar} \lambda' x}$$

$$= |A|^2 \int_{-\infty}^{\infty} e^{-\frac{i}{\hbar} (\lambda - \lambda') x} \, dx = |A|^2 2\pi\hbar \, \delta(\lambda - \lambda')$$

$$|A|^2 = \frac{1}{2\pi\hbar} \Rightarrow A = \frac{1}{\sqrt{2\pi\hbar}}$$

$$\Rightarrow \quad \langle \psi_\lambda | \psi_{\lambda'} \rangle = \delta(\lambda - \lambda')$$

for any $f(x) \in \mathcal{H}$.

$$f(x) = A \int_{-\infty}^{\infty} C(t) \, e^{\frac{it}{\hbar} x} \, dt \Leftarrow$$

$$\Rightarrow \boxed{C(P) = \int_{-\infty}^{\infty} f(x) \, e^{-\frac{iPx}{\hbar}} \, dx} \Leftarrow$$

Measurement

Back to the physical world. Say we have a state $|\psi\rangle$ which lives in the Hilbert Space.

Say you want to measure a which has the operator $\hat{A}$.

- The eigenfunctions of $\hat{A}$ are complete:

$$|\psi\rangle = c_1 |\phi_1\rangle + c_2 |\phi_2\rangle + \cdots + c_n |\phi_n\rangle.$$

where $\hat{A}|\phi_i\rangle = a_i |\phi_i\rangle$, assume $\langle\phi_i|\phi_i\rangle = 1$ (normalize)

- The eigenfunctions with different eigenvalues are orthogonal.

So $\boxed{\langle\phi_i|\phi_j\rangle = \delta_{ij}}$

Given these, let's find $\langle\hat{A}\rangle$.

$$\langle\hat{A}\rangle = \langle\psi|\hat{A}\psi\rangle.$$

- $|\hat{A}\psi\rangle = c_1 \hat{A}|\phi_1\rangle + c_2 \hat{A}|\phi_2\rangle + \cdots + c_n \hat{A}|\phi_n\rangle$

$= c_1 a_1 |\phi_1\rangle + c_2 a_2 |\phi_2\rangle + \cdots + c_n a_n |\phi_n\rangle$

• $\langle\psi| = c_1^* \langle\phi_1| + c_2^* \langle\phi_2| + \cdots + c_n^* \langle\phi_n|$

Exp Show $\langle\psi|\hat{A}\psi\rangle = \sum_{n=1} |c_n|^2 a_n$

Exp : Show $\sum |c_n| = 1$.

• Now assume we need to measure two quantites $\hat{A}, \hat{B}$. for a state $|\psi\rangle$

with

Eigenfunction of $\hat{A} = \{|\phi_1\rangle, \dots |\phi_n\rangle\}$

Eigenfunction of $\hat{B} = \{|\varphi_1\rangle, \dots |\varphi_n\rangle\}$

then

$$|\psi\rangle = c_1 |\phi_1\rangle + c_2 |\phi_2\rangle + \dots + c_n |\phi_n\rangle$$

and

$$|\psi\rangle = \gamma_1 |\varphi_1\rangle + \gamma_2 |\varphi_2\rangle + \dots + \gamma_n |\varphi_n$$

• If you measure first $\hat{A}$. You will find, say, at state $|\phi_i\rangle$ for some i.

$$|\psi\rangle \longrightarrow |\phi_i\rangle ,$$

now, we know what the value of the observable A, it is just the eigenvalue of $|\phi_i\rangle$.

• Now, we want to measure B, our state has become $|\phi_i\rangle$, but $|\phi_i\rangle$ in Hilbert Space, so

$$|\phi_i\rangle = b_1|\psi_1\rangle + b_2|\psi_2\rangle + \cdots + b_n|\psi_n\rangle$$

after you measure, you find $|\psi_j\rangle$, so

$$|\psi_j\rangle = z_1|\phi_1\rangle + z_2|\phi_2\rangle + \cdots + z_n|\phi_n\rangle$$

we miss the information that we had about the observable $\hat{A}$!!

(not always).

Stern - Gelach experiments:

let $\hat{A} := \hat{S}_z$ spin in the z-direction; with eigenfunctions $\{|\uparrow\rangle, |\downarrow\rangle\}$

$\hat{B} := \hat{S}_x$ spin in the x-direction: with eigenfunctions $\{|\rightarrow\rangle, |\leftarrow\rangle\}$